
Reproducing and Enhancing LLaGA: Random Walk Sampling and Self-Attention Projection for Graph-to-LLM Alignment

DATS369 Machine Learning with Graphs

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Abstract:

While LLMs perform well on sequential data, they are not inherently designed to process graph-structured inputs due to their non-linear and relational nature. Existing approaches, such as text-based graph descriptions or task-specific fine-tuning, often lose structural information or reduce generalizability.

LLaGA (Large Language and Graph Assistant) addresses this by converting subgraphs into token sequences through template-based sampling, embedding, and projection, enabling frozen LLMs to understand graphing structure input.

In this project, I reproduce two LLaGA baselines (Node Detailing and Hop Overview) and propose two enhancements: a Random Walk-based sequence template that captures high-frequency local neighbors, and a masked self-attention projector that allows contextual information exchange between nodes. Experiments on node classification and link prediction tasks over Cora and PubMed demonstrate that

my modifications offer a compatible and more efficient, scalable alternative to the original design.

Introduction:

Large language models (LLMs) have shown impressive capabilities in natural language understanding and reasoning. However, they struggle to directly process graph-structured data due to its non-sequential and relational nature.

Several existing methods have attempted to bridge the gap between LLMs and graph-structured data. One approach is to convert subgraphs into plain-text descriptions, but this often loses critical structural information and does not scale well. Another work, such as InstructGLM, fine-tunes LLMs using task-specific natural language prompts, but this compromises generalization and requires model-specific adaptation. Other methods like GraphGPT combine node attributes with graph neural networks, but their structural encoders may fail to produce token representations compatible with LLM input expectations.

LLaGA (Large Language and Graph Assistant) addresses this challenge by converting graph-structured data into fixed-length token sequences through template-based neighborhood serialization and a trainable projection layer, effectively transforming subgraphs into token sequences that are compatible with frozen LLMs.

In this project, we reproduce the LLaGA-Node Detailing -7B and LLaGA-Hop Overview -7B baseline and propose two enhancements. First, we introduce a **Random Walk (RW)**-based sequence template retrieving node sequence by relevance to replace the rigid, hop-wise sampling used in the original design. Second, we substitute the independent MLP projector with a **masked self-attention layer**, allowing each node to incorporate information from its neighbors. These changes aim to improve structural representation and produce lightweight and scalable architecture.

Methodology:

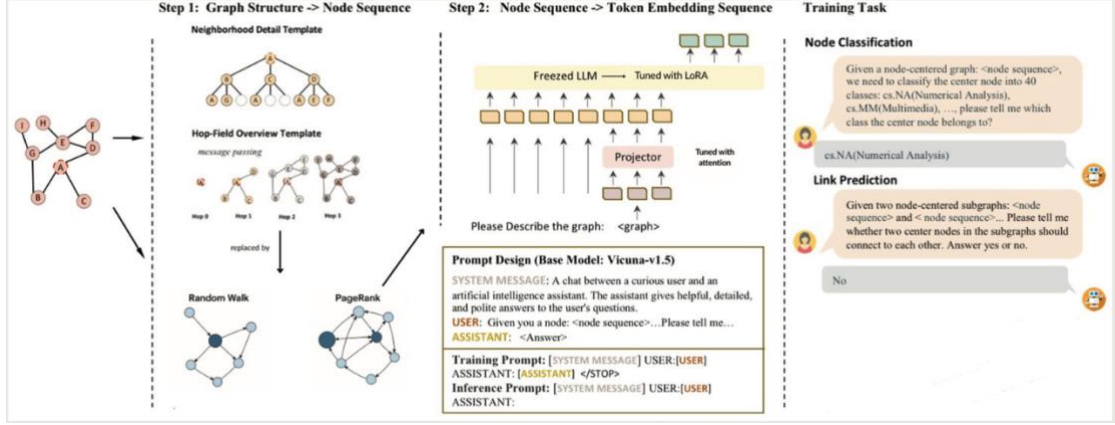


Figure 1. LLaGA Workflow

1. Random Walk-Based Sequence Template:

In the original LLaGA framework, node sequences are constructed using predefined templates such as the Neighborhood-Detail (ND) or Hop-Overview (HO) template, which rely on hop-based sampling or aggregation. To improve flexibility and structural relevance, we develop Random Walk (RW)-based approach.

For each target node v , we perform N independent random walks of length ℓ and count how frequently each node $u \in V$ is visited: $f(u) = \sum_{n=1}^N \sum_{t=1}^{\ell} I[v_t^{(n)} = u]$. where $v_t^{(n)}$ is the node visited at step t of the n th walk, and $I[\cdot]$ is the indicator function. The top K most frequently visited nodes are selected to form a fixed-length sequence: $S_v = \text{TopK}(\{u \in V | f(u)\}, K)$. This sequence is then ordered by descending visitation frequency and passed to the embedding and projector modules.

Compared to ND and HO, the RW template adapts better to the local structure of the graph. It avoids over-representing uniformly sampled nodes from low-degree regions and instead focuses on structurally or semantically related neighbors. Moreover, the RW method naturally produces compact, padding-free sequences that align well with the fixed token input size of LLMs, reducing noise and improving sequence efficiency. Since each random walk is independent, this method is also parallelizable, making it scalable to large graphs.

2. Self-Attention Projector

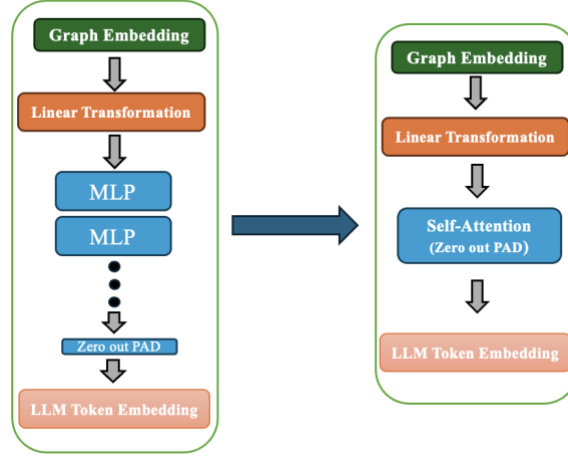


Figure 2. Replace MLP layer by Self-Attention Layer

In the original LLaGA design, the projector module consists of two parts. First, a linear transformation is applied to map the graph embeddings into the LLM’s token embedding space, ensuring dimensional alignment as well as zeroing out the embeddings of padding token so as not interfere downstream processing. Then, the multi-layer perceptron (MLP) layers are used to further refine the representations through non-linear transformations and normalization. To enhance the ability of each node to incorporate information from its neighbors, we replace the MLP component with a multi-head self-attention layer. This allows each token to attend to others in the sequence and dynamically assign attention weights based on structural relevance.

The workflow begins with a linear transformation to align the graph

embeddings $H_{graph} \in R^{L \times d_{in}}$ with the LLM’s expected input $H^{(0)} \in R^{L \times d_{model}}$:

$H^{(0)} = H_{graph}W_p + b_p$. Then, with $H^{(0)}$, we compute Query, Key, and Value

matrices using three separate learned linear layers: $Q = H^{(0)}W^Q, K = H^{(0)}W^K, V =$

$H^{(0)}W^V$. The matrices are split into h heads and passed through scaled dot-product

attention: $Attention(Q_i, K_i, V_i) = softmax(\frac{Q_i K_i^T}{\sqrt{d_k}})V_i$.

The outputs from all heads are concatenated and projected using a final linear layer to obtain the token embeddings for the frozen LLM: $MultiHead(H^{(0)}) =$

$[head_1; \dots; head_h]W^o, W^o \in R^{(hd_k) \times d_k}$.

To handle padding tokens in case we use Node-Detailing template, we first use

a (`key_padding_mask`) to set the Q-K similarity scores of padding positions to $-\infty$ during the attention computation, so their weights become zero after SoftMax. Then, after computing the final attention outputs, I explicitly set the output embeddings at all padding positions to zero, ensuring they carry no information into the LLM.

This self-attention-based projector provides two major advantages. First, it enables each node to access information from other nodes in the sequence, effectively capturing the global structure of the subgraph. Second, different heads can focus on different aspects of the graph—some attending to local neighbors, others to structurally central nodes—offering a more expressive and flexible representation. Compared to an independent MLP, this design improves alignment with the LLM’s reasoning process and leads to better task performance with minimal added computational overhead.

Supplementary Comparison – Random Walk and PageRank-Based Sampling:

Both Random Walk and PageRank rely on the stochastic traversal in graph, but RW focus on local feature, Page Rank produce global information.

Specifically, RW is a localized, lightweight sampling method. It selects frequently visited neighbors within a limited number of steps, emphasizing nodes that are closely related to the target. The walk terminates after a fixed length or when no further neighbors exist, without a restart mechanism. This makes RW suitable for capturing local structural patterns and producing compact, prompt-friendly node sequences.

In contrast, PageRank performs global information aggregation. It models the probability of landing on each node by repeatedly propagating scores across the entire graph. A damping factor introduces a small probability of randomly jumping to any node in the graph. This prevents the walk from getting trapped in dead ends or small subgraphs. The final scores reflect node centrality in the whole graph rather than relevance to a specific center node.

Experimental Setup:

Dataset - We conduct experiments on two benchmark graph datasets:

- Cora: 2,708 Node, 5429 Edges, 7 Classes, each node has a short document as textual features.
- PubMed: 19717 nodes, 44338 Edges, 3 Classes, each feature are derived from medical abstracts.

I evaluate the model on two tasks: Node Classification (NC) and Link Prediction (LP).

Frozen LLM and Encoder - My implementation is based on the official LLaGA codebase, using the Vicuna-7B language model as a frozen backend. Node textual features are encoded using the SimTeG encoder, and no fine-tuning is applied to the LLM. I reproduce both the ND and HO baselines.

Implementation Summary- To implement the Random Walk (RW) template, I added a new sampling function to the data_process.py file cloned from the official LLaGA GitHub, alongside the existing ND and HO template functions. Specifically, I set the Hyperparameter **N=10, walk length $\ell=10$, and top K=50**.

To implement the masked self-attention projector, I replace the original MLP layers with the built-in (MultiheadAttention) module from **PyTorch**, using 4 attention heads. A linear projection layer is retained to align graph embeddings to the LLM's token dimension, followed by the attention layer to enable information exchange across the sequence. Padding tokens are masked using a (key_padding_mask) during attention and are explicitly zeroed out after attention to prevent semantic interference.

Evaluation Result:

	Node Classification ACC		Link Prediction ACC	
	Cora	PubMed	Cora	PubMed
Hop-Field Overview Template- (MLP Layer)	0.8690	0.9503	0.8582	0.8918
Node Detailing- (MLP Layer)	0.8745	0.9495	0.8235	0.9143
Random Walk - (MLP Layer)	0.8727	0.9483	0.6941	0.8970

Hop-Field Overview Template - (Self-Attention Layer)	0.8635	0.9495	0.6706	0.7589
Node Detailing - (Self-Attention Layer)	0.3764	0.9058	0.6838	0.8927
Random Walk - (Self-Attention Layer)	0.8560	0.94828	0.7000	0.89698

Result Analysis:

Template comparison: RW vs. ND&HO

Comparison From the Template level:

Random Walk (RW) template performs comparably to ND and HO on PubMed, but experiences noticeable accuracy drop on **Cora** (approximately -19.18%), particularly when using the MLP projector. This discrepancy can be attributed to the differences in graph size and density. PubMed is a larger and denser graph, allowing random walks to explore a broader and more representative neighborhood. In contrast, Cora's smaller size and node degrees means that walks often revisit a limited set of nearby nodes, resulting in more localized sequences.

Comparison from Task Level:

Link prediction (LP) accuracy tends to drop more than node classification (NC) when using the RW template (approximately). This potentially because NC primarily relies on local node features, which RW captures well through frequent neighbor sampling. However, LP requires accurate modeling of edge-level relationships. RW may overlook infrequent but structurally important nodes or edges, potentially breaking connectivity patterns that LP depends on.

Projector Comparison: Self-Attention vs. MLP

Comparison from dataset level:

On the Cora dataset, we can observe a larger drop in evaluation accuracy when using self-attention. This is likely due to the smaller size and higher sparsity of Cora, where node sequences contain many padding tokens. These [PAD] entries dilute attention weights, reducing the effectiveness of contextual aggregation.

Comparison from Task Level:

the link prediction (LP) task drops more than node classification (NC) with the self-attention layer. I consider the potential reason that NC benefits from general feature fusion, while LP requires precise modeling of edge-level relationships. Self-attention may allow both connected and unconnected nodes to absorb similar neighboring information, thereby blurring the distinction between edge-specific features.

Conclusion:

In this project, I reproduced two LLaGA baselines—Node Detailing (ND) and Hop Overview (HO)—with results closely matching the original paper and proposed two enhancements to improve graph-to-LLM alignment: a Random Walk (RW)-based sequence template and a masked self-attention projector. My experiments on node classification and link prediction tasks across two benchmark datasets, Cora and PubMed, show that RW offers a lightweight and localized alternative to fixed-hop sampling, performing particularly well on larger graphs. Meanwhile, the self-attention projector enables richer contextual interactions among nodes, though its effectiveness can be reduced by excessive padding in smaller graphs or by diluting edge-specific signals in link prediction tasks.

These results demonstrate that modifying the sequence construction and projection stages in LLaGA can lead to improved performance and scalability, especially for graph structures that are large or sparse.

References

- [1] Chen, Y., Wang, R., Zhang, Y., Shi, Y., Shao, Y., Zhao, D., & Yan, R. (2024). LLaGA: Large Language and Graph Assistant. arXiv preprint arXiv:2402.08170.